Carnegie Mellon University
Computer Science Academy

Design and Development Guide

Before beginning this project, be sure to review the rubric to ensure you include all the required elements.

Description

Describe your program. What do you envision your app will do?

Note: At this step, do not be concerned about how to do it, just focus on what you envision.

My program will allow players to play a bounce ball game. In the game, players will need to move the bouncing ball to avoid obstacles and arrive at the final location for each level of the game. Players will be able to move the ball by pressing the keys: w, s, a, d, which means that the ball will move up, down, left, and right. Also, players will be able to move on only the safe blocks, which will be shaped like rectangle. The obstacles for each level will move continuously until the player finished the level.

There will be total three different levels: each level will have different styles of obstacles requiring players to move differently from the other levels. The final location will be marked as a crown, and the players will need to hit the crown in each round.
As soon as players hit with the star, players will play the next level of the game in which they will interact different settings: different shapes of obstacles and different map settings. After you finished all the three levels, you will receive a message from the game that says you are the winner now.

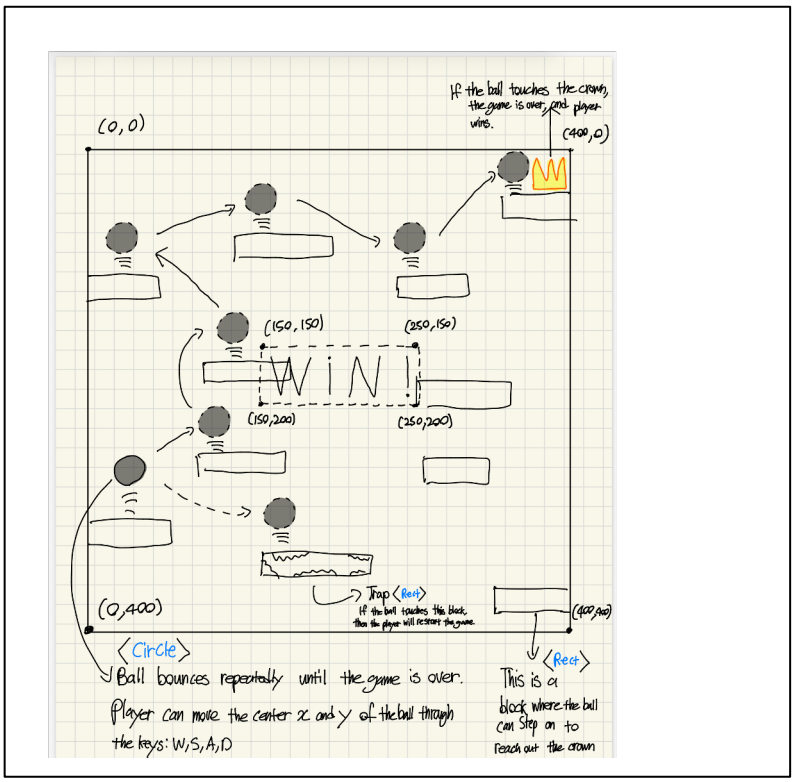
As a warning, if you ever touch the obstacles or fall from the safe blocks, where these are the only safe places that the ball can move, the bouncing ball will be re-positioned at the starting point and will need to re-play the game.

Canvas Design

Sketch your canvas. Label any animation or user interaction.

- **Colors:** What colors will be dominant in your app? Are there specific shades that are important to the app's functionality or aesthetic?
- **Animations/Movements:** Will there be any animated elements? If yes, describe how you envision them. For example, will a shape move across the screen?

(0,0)



(400,400)

Code Design

Plan your code. List here the coding concepts and properties you plan to use to accomplish your goals, and why you plan to use them. These should incorporate at least two new concepts from this unit. Briefly describe what purpose each element will serve in your project. Aim for at least four to six properties.

Concept	Purpose
Groups (unit 6)	I plan on using Groups to create map settings more efficiently. Each level will have different map settings in the game. Instead of creating every part of map separately, Groups will help organize the codes clearer and will be flexible when I need to change the level of the game easily.
Motion (unit 6)	I first plan on using Motion to make the default movement of the ball during the game. Since the game is intended to be a bounce ball game, the ball has to bounce continuously unless the game is over, in which the Motion can make the ball bounce. I second plan on using Motion to make the movement of the obstacles that need to keep interrupt players from controlling the bouncing ball.
Conditionals (unit 3)	I plan on using Conditionals to create some cases that would happen during the game. For example, if the ball fell off below the window (under 400), then the game will re-start. Another, if the ball hit the obstacles, then the game also will re-start. Lastly, if the ball hit the crown, then the player will move on to the next level of the game.

Onkeypress (unit 4)	I plan on using onkeypress to make movements of the ball: up, down, left, and right. Therefore, players will be able to move the bouncing ball in those directions to avoid the obstacles during the game by pressing those keys.
Shapes method (unit 4) (hitsShape, app.stop)	I plan on using hitsShape to make a rule that runs if the ball hits the obstacles during the game. If the ball hits the obstacles, then the player will re-start the game at the starting point. I plan on using app.stop to make an ending for every interaction with obstacles during the game. If the player touches with the obstacles, the ball and the other things in the game will stop moving. Then, the ball will be re-positioned at the starting point.
onMousePress (unit 2)	I plan on using onMousePress to allow players to pick the level of the game themselves by using the code "onMousePress." Players will have a time to decide which level they are going to play by clicking on level labels.
onMouseMove (unit 3)	Similar as I said above, I plan on using onMouseMove to allow players to pick the level of the game themselves by moving the cursor in the window by using the code "onMouseMove."

Creative Task Reflection

After completing your creative task use this form to reflect on your work.

Difficulties and Opportunities

For this section, go in-depth into either a difficulty you faced or an opportunity you saw to improve or add to your code.

If you Faced a Difficulty:

- Describe the Problem: What specific issue did you encounter?
- Attempted Solutions: What did you try in order to solve the problem?
- Final Solution: What was the final solution that fixed the issue?

If you Found an Opportunity:

- Reason for the Addition/Change: What made you consider this opportunity? Was it a challenge or an aha-moment?
- Impact on Design: How did seizing this opportunity impact other aspects of your design?
- Design Trade-offs: What aspects of your original design did you have to adjust or remove to incorporate this new feature?

I faced a difficulty. Since I created three map settings, I had to separate and remove two of them while playing on one of them. However, I only thought making them invisible will work well. As a result, while I played one of them, the invisible map setting of other levels did not allow me to play one particular map field. I tried two different ways, which were making them invisible and using g.remove() code. As said earlier, invisible did not work because the map setting of other levels still existed. On the other hand, g.remove() code was a different issue. I had to put one shape by one into the code from a group. As I put too many shapes at the same time, the code got an error. Therefore, my final decision for this problem

was to use the invisible code.

Finished Project Code Outline

List **only one instance** of each important coding concept used in your project, along with the corresponding line number, and describe how it contributes to the overall function of your program. **Highlight the aspect you are most proud of.** For each concept, also include a brief definition to demonstrate your understanding. By focusing on one instance of each concept, aim to provide more depth and clarity in your responses.

Concept	Line number(s)	Definition	Purpose
Groups	10-17, 22-41, 51-68, 85-94, 100-108	A code that allows us to treat multiple shapes as a single shape	I planned on using Groups to create map settings more efficiently. Each level has different map settings in the game. Instead of creating every part of map separately, Groups helped organize the codes clearer and was flexible when I need to change the level of the game easily.
Motion	127-178	A code that animates a shape, so it appears to be moving.	I first planned on using Motion to make the default movement of the ball during the game. Since the game is intended to be a bounce ball game, the ball has to bounce continuously unless the game is over, in which the Motion can make the ball bounce. I second planned on using Motion to make the movement of the obstacles that need to keep interrupt players from controlling the bouncing ball.
Conditionals	132-178,187-192, 200-205, 212	A code that will run if a statement is true, useful to verify situations	I planned on using Conditionals to create some cases that would happen during the game. For example, if the ball fell off below the window (under 400), then the game re-starts. Another, if the ball hit the obstacles, then the game also re-starts. Lastly, if the ball hit the crown, then the player moves on to the next level of the game.

Onkeypress	186	A code that is called when a key is pressed	I plan on using onkeypress to make movements of the ball: up, down, left, and right. Therefore, players is able to move the bouncing ball in those directions to avoid the obstacles during the game by pressing those keys.
Shapes method	132-138,162-178, 191, 204, 211	shape1.hitsShape(shape 2) tests if a shape hits another shape	I plan on using hitsShape to make a rule that runs if the ball hits the obstacles during the game. If the ball hits the obstacles, then the player re-starts the game at the starting point.
onMousePress	211	A code that will run if the mouse is pressed	I plan on using onMousePress to allow players to pick the level of the game themselves by using the code "onMousePress." Players have a time to decide which level they are going to play by clicking on level labels.
onMouseMove	122	A code that will run as the mouse moves	Similar as I said above, I plan on using onMouseMove to allow players to pick the level of the game themselves by moving the cursor in the window by using the code "onMouseMove."

Future goals

Reflect on your entire design and coding process, from conceptualization to implementation. In this section, consider not only what you would change in the final product but also how you would approach the entire project differently.

- **Code Changes:** What specific changes would you make to your code if you had more time or if you were to start from scratch? Would you add new features or refine existing ones?
- **Process Adjustments:** Did you stick to the original plan, or were there deviations? Were these changes beneficial or did they lead to complications? How could your planning or approach be improved?
- **Resource Utilization:** Did you make the best use of available resources, like class notes, tutorials, or peer help? What additional resources could have been beneficial?
- **Time Management:** Reflect on how effectively you managed your time. What would you do differently to ensure a smoother process?

By considering these aspects, you will not only think about 'what' you would change but also 'how' and 'why,' which can be incredibly enlightening for future projects.

What I want to change is the process of my project. Since I tried to design everything even before to start working on coding, I had struggled to imagine every detail of the game in my brain. Therefore, it was challenging to transfer my thought to the coding language directly. I often faced some issues with codes while running my programming. Therefore, for the next time, I decided to be more brave starting coding before talking about how I am going to present the project.